DC-DC BOOST CONVERTER PROTOTYPE

For PV-Integrated Electric Vehicle (Scale-Down Prototype)

20W Solar Panel | 48V Battery Output | DC-DC Boost Converter Design
week 07

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SOLUTION TIME STEP

In PSCAD, the solver works by taking snapshots of the circuit at fixed time intervals called the solution time step. It can only switch the MOSFET ON or OFF at these exact snapshot moments.

For a switching frequency of 20 kHz,

switching period $T = 50\ \mu s$

For a duty cycle of $D = 0.6265$,

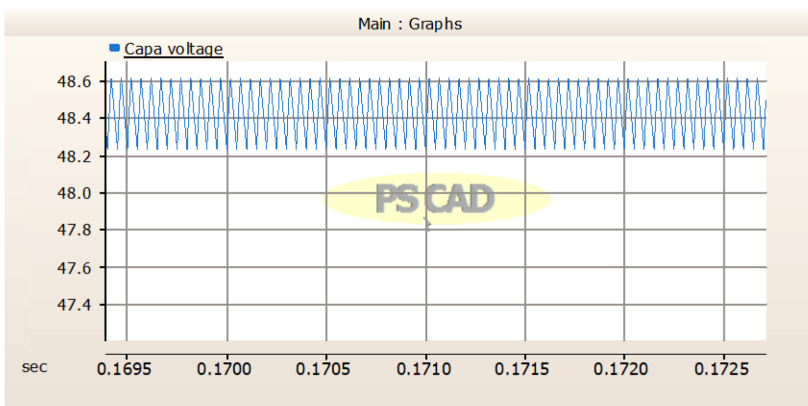
Required ON-time = 0.6265×50

$= 31.325\ \mu s$.

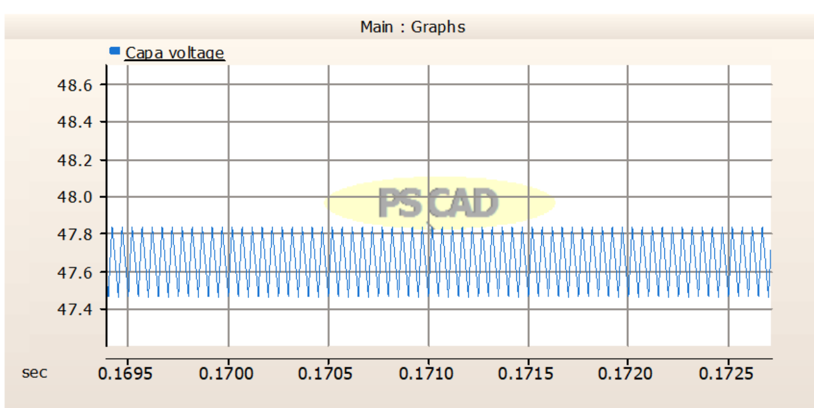
With the original solution time step of 0.5 µs, the solver could not switch at exactly 31.325 µs — it rounded to the nearest available snapshot, resulting a small error in the actual duty cycle applied.

Since the boost converter output voltage is highly sensitive to duty cycle , even this small rounding error caused the output voltage to deviate from the 48 V target.

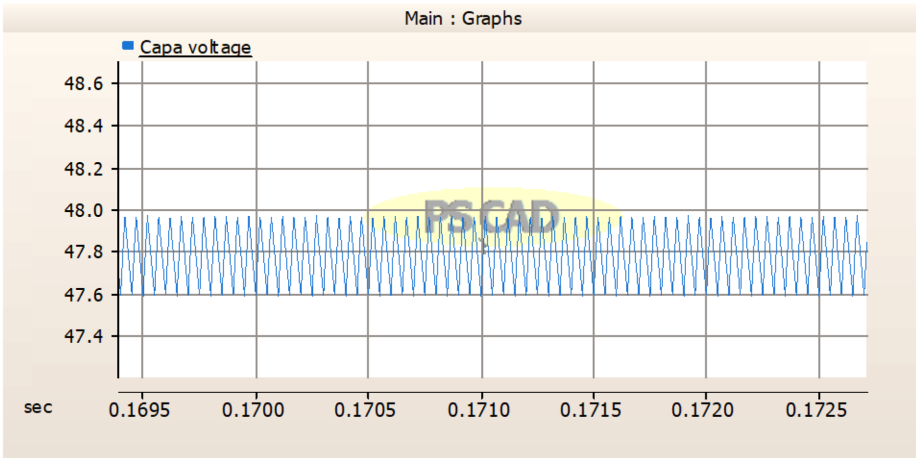
By reducing the solution time step to 0.315 µs, the solver gained finer resolution and could represent the switching instant much more accurately, bringing the simulated output voltage to approximately 48 V as expected from the theoretical design.



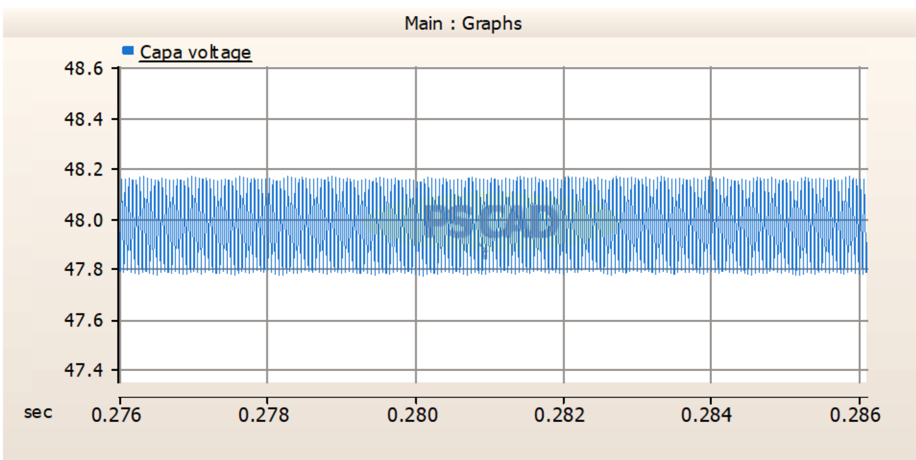
0.2



0.5



0.3125



0.315

Time Step (µs)	Avg Vout (V)	Vout Ripple ΔV (V)	Ripple %
Designed value for 33uF	48.00	0.396	0.82%
0.200 µs	47.65	0.30	0.63%
0.300 µs	47.80	0.33	0.69%
0.3125 µs	48.43	0.30	0.62%
0.315 µs	47.97	0.38	0.79%

CURRENT RIPPLE AND INDUCTOR

- At 30% ripple ,

$$\Delta I_L = 0.30 \times 1.115 = 0.3345 \text{ A}$$

The minimum inductance required to maintain Continuous Conduction Mode (CCM) is:

$$L_{min} = \frac{V_{in(min)} \cdot D_{max}}{\Delta I_L \cdot f_{sw}}$$

$$L_{min} = \frac{17.93 \times 0.6265}{0.3345 \times 20000}$$

$$= 1677 \ \mu\text{H}$$

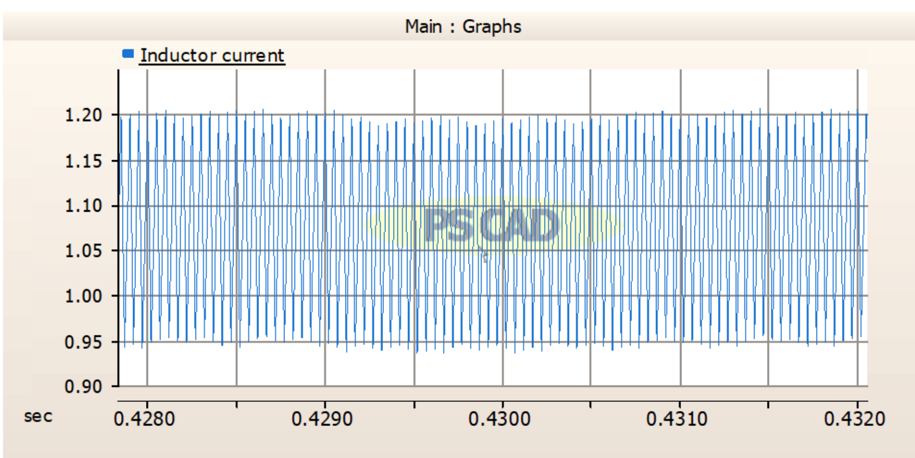
- At 20% ripple , $\Delta I_L = 0.20 \times 1.115 = 0.223$ A

$$L_{max} = \frac{V_{in(min)} \cdot D_{max}}{\Delta I_L \cdot f_{sw}}$$

$$L_{max} = \frac{17.93 \times 0.6265}{0.223 \times 20000}$$

$$= 2519 \ \mu\text{H}$$

- Inductance range: 1677 μH to 2519 μH



COMPONENT LIST

Component	Selected Part	Specification / Reason
Solar Panel	20W Poly/Monocrystalline	V_{mp}=17.93V, I_{mp}=1.11A, P=20W
MOSFET (Switch) Diode	IRF540N MBR10100CT Schottky	Rated 100V/28 A, low Rds(on) 20A / 100V, fast switching, low V_f
Inductor	1.7 mH Toroidal	Rated ≥2A, ferrite core, ensures CCM
Output Capacitor	33 µF / 100V Electrolytic	Smooth 48V output, 100V safety margin
Gate Driver	IR2110	Drives MOSFET gate from MCU PWM signal
PWM Controller	Arduino Uno	Generates 20 kHz PWM at 62.65% duty cycle
Load Resistor (Lab)	120 Ω / 25W Wirewound	Initial testing before battery connection